

MOCH AFIFUL ISLAM

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Education

Universitas Brawijaya

Bachelor of Mathematics (GPA: 3.18/4.00)

Malang, Indonesia

Expected August 2026

Relevant Coursework: Digital Image Processing, Machine Learning, Computer Vision, Numerical Optimization, Numerical Methods, Operation Research, Linear Algebra, Calculus, Probability and Statistics, Data Structures and Programming Algorithms.

Experience

National Research and Innovation Agency (BRIN)

Computer Vision Researcher

Bandung, Indonesia

February – July 2026

- Developed object detection and computer vision models using Python, PyTorch, and OpenCV for scientific research and real-world datasets.
- Designed end-to-end machine learning pipelines, including dataset collection, annotation, preprocessing, model training, validation, and hyperparameter optimization.
- Evaluated model performance using Precision, Recall, F1-Score, and confusion matrix to ensure reliable and reproducible results.
- Collaborated with multidisciplinary researchers to investigate AI-based solutions for scientific and industrial applications.
- Prepared technical documentation, experimental reports, research manuscripts, and presentation materials following reproducible research practices.
- Conducted literature reviews and experimental analyses to identify state-of-the-art approaches in computer vision research.

Mathematics Department of Universitas Brawijaya

Teaching Assistant

Malang, Indonesia

February – December 2025

- Assisted undergraduate courses in Programming Algorithm II, Numerical Methods, Numerical Optimization, and Digital Image Processing through laboratory instruction and academic mentoring.
- Guided more than 100 undergraduate students in laboratory sessions covering programming, numerical computation, optimization algorithms, and digital image processing techniques.
- Demonstrated practical implementation of computational algorithms using Python, MATLAB, and OpenCV to bridge theoretical concepts with real-world applications.
- Prepared laboratory modules, project assignments, practical examinations, and instructional materials in collaboration with course lecturers.
- Mentored students in debugging, algorithm implementation, and computational problem-solving to improve programming proficiency.

PT Bank Syariah Indonesia Tbk.

Data Analytics Intern

Batam, Indonesia

January – February 2025

- Developed a K-Means clustering-based customer segmentation project by leveraging banking operational data to identify customer groups and support strategic marketing decisions.
- Managed administrative documentation and operational records to support daily banking activities and maintain data accuracy.
- Verified customer documentation, financial records, and operational transactions in compliance with banking procedures and internal policies.
- Processed, validated, and analyzed operational and transactional datasets to support internal reporting and performance monitoring.
- Collaborated with branch personnel to maintain efficient administrative workflows, ensure documentation accuracy, and support operational excellence.

Projects

Deep Ensemble Learning for Chili Variety Identification

Bandung, Indonesia

Machine Learning Project

March – June 2026

- Developed a deep ensemble learning framework for automatic chili variety classification.
- Compared transfer learning and stacking classifier to improve classification accuracy.
- Accepted for presentation at international conference on computer, control, and its application (IC3INA) 2026.

YOLOv12 for Motor Vehicle Detection and Counting on CCTV Recordings

Malang, Indonesia

Undergraduate Thesis

September 2025 – February 2026

- Developed an intelligent traffic monitoring system using YOLOv12 for real-time motor vehicle detection and counting.
- Constructed and annotated a custom CCTV vehicle dataset collected from multiple intersections in Malang City.
- Implemented transfer learning using the Ultralytics YOLO framework to improve detection performance.
- Integrated object detection with DeepSORT for multi-object tracking and vehicle counting.
- Evaluated model performance using Precision, Recall, and mean Average Precision (mAP).

Skin Cancer Classification Using VGG19

Malang, Indonesia

Computer Vision Project

February – June 2025

- Developed a deep learning model using the VGG19 architecture for automated classification of seven skin lesion categories from dermoscopic images.
- Applied image preprocessing, data augmentation, and transfer learning to improve model generalization and classification performance.
- Demonstrated the potential of artificial intelligence to support computer-aided medical diagnosis through image classification.

Customer Segmentation for Banking Using K-Means Clustering

Batam, Indonesia

Internship Project

January – February 2025

- Developed a customer segmentation model to identify distinct customer profiles based on banking transaction and financial behavior data.
- Performed data preprocessing, feature engineering, and exploratory data analysis to improve clustering quality and interpretability.
- Evaluated clustering performance using PCA and silhouette score to support customer profiling and business decision-making.

Leadership & Organization

FORKALAM FSTeM Universitas Brawijaya

Malang, Indonesia

Chairman

January – December 2025

- Led an Islamic student organization with more than 60 active members across eight departments.
- Directed the implementation of more than 50 religious, social, and self-development programs serving over 90 students within the Faculty of Mathematics and Natural Sciences.
- Managed organizational strategic planning, budget allocation, and annual performance evaluation.

FORKALAM FSTeM Universitas Brawijaya

Malang, Indonesia

Deputy Chairman

January – December 2024

- Assisted the Chairman in coordinating organizational activities involving more than 50 active members.
- Supervised cross-department coordination to improve operational effectiveness.
- Monitored the implementation of religious, educational, and community engagement programs.

FORKALAM FSTeM Universitas Brawijaya

Malang, Indonesia

Staff of Mosque and Al-Qur'an Department

January – December 2023

- Organized Islamic studies, Qur'an recitation, and student spiritual development programs.
- Assisted in planning and coordinating faculty-level religious events.
- Collaborated with organizational members to strengthen student engagement in campus activities.

Volunteering

FSTeM Universitas Brawijaya

Malang, Indonesia

Deputy Village Coordinator of Community Service Program (KKN)

June – July 2024

- Coordinated community service initiatives focusing on public health, environmental awareness, and waste management.
- Collaborated with village leaders and local residents to plan and implement community development programs.
- Assisted the Village Coordinator in managing schedules, team communication, and resource allocation for multidisciplinary student teams.

National Achievement Center (Puspresnas)

Malang, Indonesia

Arabic Debate Committee of MTQMN 2023

August – November 2023

- Coordinated the technical implementation of the National Arabic Debate Competition involving participants from universities across Indonesia.
- Managed competition schedules, coordinated with judges, and ensured compliance with national competition procedures.
- Communicated with more than 50 participants and officials throughout the competition.

FORKALAM FSTeM Universitas Brawijaya

Malang, Indonesia

Public Relations & Fundraising Division of Ramadhan MIPA 2023

March – April 2023

- Led public relations and fundraising activities for a faculty-wide Ramadan program involving more than 80 participants.
- Secured more than IDR 9 million in sponsorships to support event implementation.
- Established partnerships with internal organizations and external stakeholders to strengthen collaboration and event success.

Honors & Awards

Musabaqah Hifdzil Qur'an 20 Juz (MHQ20) MTQ Universitas Brawijaya

Malang, Indonesia

2nd Place (University-level Competition)

May 2025

Musabaqah Fahmil Qur'an (MFQ) MTQ Universitas Brawijaya

Malang, Indonesia

1st Place (University-level Competition)

June 2024

Musabaqah Debat Ilmiah Al-Qur'an (MDIA) MTQ Universitas Brawijaya

Malang, Indonesia

2nd Place (University-level Competition)

May 2023

Technical Skills

- **Programming:** Python (Advanced) and MATLAB (Intermediate).
- **Artificial Intelligence:** Computer Vision, Deep Learning, Classification, Object Detection, Multi-Object Tracking, Image Preprocessing, and Transfer Learning.
- **Frameworks:** PyTorch, OpenCV, Ultralytics YOLO, NumPy, Pandas, Scikit-learn, and Matplotlib.
- **Tools:** Visual Studio Code, Google Colab, Git, GitHub, Roboflow, Kaggle, Microsoft Office, and L^AT_EX.
- **Professional Skills:** Leadership, Problem Solving, Critical Thinking, Communication, Team Collaboration, Mentoring, and Public Speaking.
- **Languages:** Indonesian (Native), English (Limited Working Proficiency), and Arabic (Professional Working Proficiency).
- **Certifications:** Generative Artificial Intelligence, Microsoft Certified: Azure AI Fundamentals (AI-900), ASEAN Data Science Explorers 2024 – Enablement Session, Microsoft Office Desktop Application, Brawijaya English Proficiency Test.